James Bray - 220123839

Predicting Household Energy Consumption with Machine Learning

Evaluating smart meters and weather data as predictors

Background

Due to wind power's variability, the UK's transition to renewable energy, particularly North Sea wind farms, has introduced grid stability challenges. Demand management is critical to balance supply during low-wind periods or peak household consumption.

Figure 1 - Standard smart meter

Two primary data streams can be used for predictions:

- **Smart meter appliance data:** Offers direct appliance level insights but suffers from low adoption rates.
- **Public weather data:** Widely available and scalable for regional demand estimation, but lacks granularity.

This investigation evaluates both approaches to develop machine learning models that enhance prediction accuracy.

The data provided is **'household_energy_data.csv'** [1] and contains the power consumed by various appliances in kW along with standard weather data. There are 50380 entries, which should be sufficient for this report.

However, the dataset lacks temporal features like timestamps, hindering analysis of daily or seasonal demand patterns. Also, this data is for one household, introducing a sampling bias. This data is less useful for grid planning than aggregated regional trends, and future work should use system-level forecasting with time-aggregated data.

Approach

It is assumed that due to the units of the data provided from the smart meter being in kilowatts, there will be a linear relationship between power consumed and power required. Therefore, linear regression should capture the trend effectively. The same cannot be said for the nonlinear weather data, and a more complex model will need to be developed to capture the dynamics of that data. These are the models to be trained in this study:

1. **Linear Regression:** This model shall be trained on the data that correlates most with the target (power consumption). This is likely to be from the smart meter readings.
2. **Linear Regression with Cross-Validation:** To validate the first model, k-fold cross-validation shall be used, which eliminates the risk of overfitting to the data
3. **Feedforward Neural Network:** To capture the trends found in the weather data, a neural network shall be trained with holdout validation. Certain hyperparameters will be adjusted to ensure the model represents the data in the best way possible.

A comparison of these two models and their respective training data will reveal how useful the sources are, potentially highlighting a new perspective on current data collection.

Data Validation

Before training a model, the input data needs to be numerical, normalised and complete. The following steps were taken to clean the data ready for training [2]:

- Remove rows that are duplicates, contain NaN, Inf, or missing values
- Remove all constant or non-numerical columns from the dataset
- Normalise the predictors to a mean of 0 and a standard deviation of 1

Data is normalised as machine learning models are sensitive to the scale of the input features, and therefore, large values will impact the model more. Normalisation eliminates this effect. These processes yield [Table 1], a sample of the standardised predictors with the desired characteristics. The target is unnormalised to maintain accuracy. See **'validate_data.m'** [1].

	NumMissing	Min	Median	Max	Mean	Std
CentralHeating_kW_	0	-0.8860	-0.2717	11.2793	-7.6882e-18	1
Dishwasher_kW_	0	-0.5568	-0.1975	7.9138	-1.4706e-17	1
UnderFloorHeater_kW_	0	-0.9977	-0.3434	10.5602	-1.3274e-16	1

Table 1 - Summary of the top three predictors after normalisation

Data Selection

Collinear data can pose significant challenges when solving linear regression, often leading to poor convergence. To address this, a preprocessing function called **'reduce_predictors.m'** [1] was implemented, which removed the following collinear features based on a correlation matrix:

`temperature ≈ apparentTemperature ≈ dewPoint, precipProbability ≈ rain`

While the dataset is not large enough for linear regression to suffer from computational limitations, eliminating redundant, noisier features becomes increasingly important for preventing overfitting. To further refine the predictor set, an additional preprocessing step using Recursive Feature Elimination (RFE) [3] was used in the **'extract_predictors.m'** [1] function.

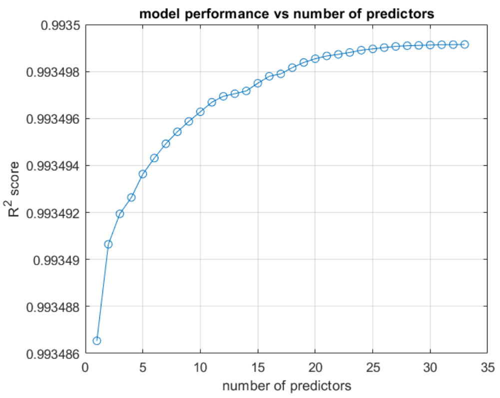

Figure 2 - Model performance vs number of features

RFE iteratively fits a linear regression model using the normal equation, identifies the least significant predictor based on its p-value, removes it, and repeats this process until only the most relevant features remain.

[Figure 2] shows diminishing returns as less important features are added. A cutoff of 20 predictors was chosen to balance simplicity and performance, avoiding overfitting while keeping the strongest contributors. See **'predictor_contribution.m'** [1]

These are the top 5 predictors of the required energy after utilising RFE:

`CentralHeating_kW_, Dishwasher_kW_, UnderFloorHeater_kW_, ElectricRadiator_kW_, Fridge_kW_`

As predicted, the smart meter readings demonstrate the best correlation with power consumed because of the incredibly high R^2 value in [Figure 2]. However, without a train/test split, there is no insight into the model's robustness. Therefore, a cross-validated model shall be explored with better generalisability and robustness.

Cross-Validated Linear Regression

K-fold cross-validation is one of the most effective methods for assessing the performance of linear regression models, especially when concerns about overfitting exist. It involves partitioning the dataset into K subsets (or "folds") and performing training K times, each time using a different fold for validation and the remaining K-1 folds for training. The final model performance is averaged over all the iterations, this is called ensemble learning. [3]

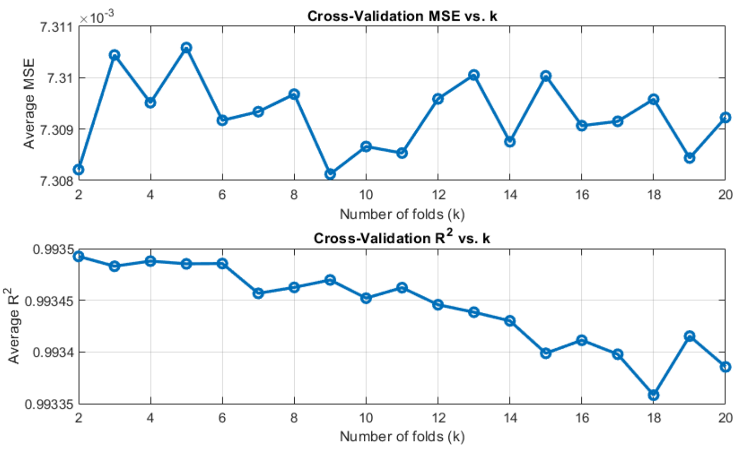

Figure 3 - Performance metrics of different fold counts

K can vary depending on the dataset size and the specific goals of the analysis. A study was done to compare the MSE and R^2 values over different K values. [Figure 3] demonstrates that the results were relatively stochastic regarding MSE, but the R^2 metric revealed that a lower K was better at fitting the data. A K value of 6 was chosen for this dataset. See **'best_k_value.m'** [1]

6 linear models were trained using the k-fold method, and the best model was selected. The function **'k_fold_linear.m'** [1] displays the metrics of each fold as it completes the training. Model 3 was selected from the set [Table 2] due to its low MSE and high R^2 value. This model is extremely accurate with very low bias, which validates that overfitting is not a significant issue with a linear model for this dataset.

fold	mse	r2
1	0.0074	0.9939
2	0.0073	0.9930
3	0.0072	0.9937
4	0.0072	0.9926
5	0.0074	0.9939
6	0.0073	0.9936

Table 2 - Metrics of each fold

Full Dataset vs Cross-Validated Model

In **'main_run.m'** [1], two methods (full dataset and cross validation) were used to generate linear models that were used to predict values based on the features in the dataset. The predictions and targets for each model were compared and evaluated using different metrics in **'get_metrics.m'** [1]. Then these metrics were compared using **'compare_metrics.m'** [1].

metric	k-fold	full	best
MSE	0.007303	0.007303	full
RMSE	0.085460	0.085457	full
MAE	0.066315	0.066315	k-fold
MAPE	13.366822	13.372200	k-fold
R2	0.993498	0.993499	full

Table 3 - Comparison of 2 linear models

Metrics in [Table 3] show that the full data model slightly outperforms the k-fold model. However, the close metrics indicate a strong linear relationship with low variance and stable predictive behaviour. Even though the k-fold model was outperformed, the analysis was needed to ensure a robust model for OOS data.

The models' near-perfect R^2 (~0.9935) confirms linear regression is ideal for this task, as the power consumption data follows a strong, predictable trend with minimal noise. Such high accuracy ensures reliable dynamic power forecasting, enabling optimised energy use without complex tuning or feature engineering.

Advanced 'No Smart Meter' Model

While the linear regression model achieved near-perfect results, this model will attempt to exclusively use external weather-related features to simulate a more realistic and scalable prediction scenario. Due to the expected nonlinear relationship between weather and energy usage, a neural network was employed for this task.

Further Data Validation

The weather data was previously neglected in the linear model due to its non-numerical type, but it is needed for this model, therefore, it is encoded into numerical data using One-hot Encoding [2]. This creates multiple new columns using the values in the variable and assigns binary values 0 or 1. This is utilised in an updated version of **'validate_data.m'** [1]. A new csv called **'weather_data.csv'** [1] was created, and the new model is trained on that.

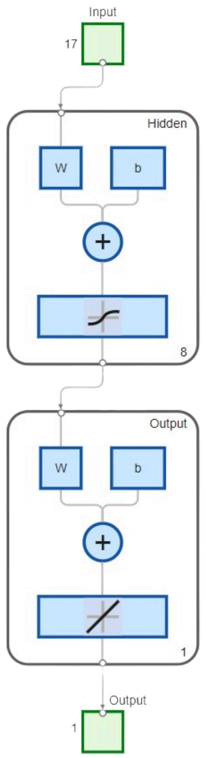

Figure 4 - Network diagram

Designing the Model

A feedforward neural network (FNN) was chosen for this task due to the nonlinear features. The **'fitnet'** [5] function was used in **'holdout_fnn.m'** [1] with the following parameters:

- **1-2 hidden layers:** Given the moderate size of the dataset and problem complexity, a small number of hidden layers is sufficient to capture nonlinear relationships without overfitting.
- **'trainlm'** [5]**:** This is a fast and efficient algorithm for small to medium-sized networks (ideal for this coursework's scale). It combines gradient descent and Gauss-Newton methods, making it suitable for regression tasks like energy prediction.
- **'tansig'** [5]**:** This is chosen for the hidden layer because it introduces nonlinearity, enabling the network to model complex relationships between weather and energy demand.
- **'purelin'** [5]**:** This is used for the output layer because the target (energy demand) is a continuous variable, and linear activation is standard for regression problems.

The **'fitnet'** [5] function utilises a holdout method, which is sufficient to neglect cross-validation. The number of neurons will now be selected by a tuning process to find the balance between model bias and variance.

Tuning Hyperparameters

There are 8 neurons in [Figure 4], however, this is a hyperparameter chosen to be variable to assess the bias-variance trade-off, as it represents the complexity of the model. [Figure 5] shows the R^2 values of models with neuron counts 2-20. Due to the incredibly low R^2 value (~0.12 max), multiple attempts at reconfiguring the model were made, including multiple layers, different activation functions, etc., but [Figure 5] shows the best attempt.

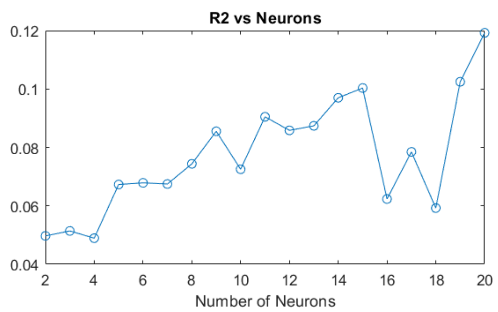

Figure 5 - R^2 vs neuron count

Linear Regression vs FNN

Linear regression achieved near-perfect accuracy ($R^2 > 0.99$) [Figure 3] with appliance data, confirming that power readings directly predict total consumption. The k-fold validation (MSE ≈ 0.0073 across folds) further validated robustness.

However, the FNN's poor performance ($R^2 \approx 0.12$) [Figure 3] suggests that weather variables alone lack predictive power for household-level demand. This aligns with the physical intuition that weather affects aggregate grid demand (e.g. regional HVAC use) but not individual homes. Perhaps more data could prove valuable when training future models.

Hybrid approaches (using weather + aggregated smart meter data) may bridge the gap between household and regional data, and should be explored when more data is available.

Conclusions

This project highlights key lessons about the use of machine learning (ML) for energy consumption forecasting. The motivation behind the project was to explore how accessible data sources, such as weather information and smart meters, could help optimise energy use through AI technologies.

However, the study revealed ethical challenges, particularly around accountability and transparency. Machine learning models often act as 'black boxes,' making it easy to overtrust poor models without rigorous validation. As AI adoption increases in society, it is critical to openly acknowledge model limitations to ensure responsible use.

Data privacy is another major ethical concern. Even with limited data availability, respecting privacy laws must remain a priority, and profit incentives should never justify breaching user rights.

Looking ahead, future challenges include improving model reliability by ensuring better data quality, fostering model interpretability, and scaling the adoption of smart technologies like smart meters. Addressing these challenges will be essential for AI to play a meaningful role in the future of energy management.

References

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